THE FIRST FACTOR OF A COMPOSITE OF FINITE TYPE IS OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The definition of a morphism of algebraic stacks being locally of finite type through smooth covers by schemes, and the stability of surjective smooth morphisms under composition and base change. The author's section on morphisms of finite type, in which the counted statement occurs. Allows the proof to read the property off the level of the covering schemes.
- The corresponding permanence statement for morphisms of schemes. Invoked inside the counted proof at source lines 3630-3631 as a cross-chapter pointer. Gives the conclusion for the middle morphism of schemes, which is what the definition requires.

Source reference. Original reference morphisms-lemma-permanence-finite-type: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3631–3631. Navigation pointer only: the counted text refers to the Stacks item morphisms-lemma-permanence-finite-type, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$, $g : \mathcal{Y} \to \mathcal{Z}$ be morphisms of algebraic stacks. If $g \circ f : \mathcal{X} \to \mathcal{Z}$ is locally of finite type, then $f : \mathcal{X} \to \mathcal{Y}$ is locally of finite type.*

Proof. We can find a diagram

$$\begin{array}{ccccc} U & \longrightarrow & V & \longrightarrow & W \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{X} & \longrightarrow & \mathcal{Y} & \longrightarrow & \mathcal{Z} \end{array}$$

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where U, V, W are schemes, the vertical arrow $W \to \mathcal{Z}$ is surjective and smooth, the arrow $V \to \mathcal{Y} \times_{\mathcal{Z}} W$ is surjective and smooth, and the arrow $U \to \mathcal{X} \times_{\mathcal{Y}} V$ is surjective and smooth. Then also $U \to \mathcal{X} \times_{\mathcal{Z}} V$ is surjective and smooth (as a composition of a surjective and smooth morphism with a base change of such). By definition we see that $U \to W$ is locally of finite type. Hence $U \to V$ is locally of finite type by Morphisms, Lemma morphisms-lemma-permanence-finite-type which in turn means (by definition) that $\mathcal{X} \to \mathcal{Y}$ is locally of finite type. $\square$